

Organic Chemistry - Functional Groups and Naming

Course: Chemistry

Learning Goals

- Recognize common organic functional groups.
- Apply basic IUPAC naming rules.
- Relate molecular structure to physical properties.

Hydrocarbons

Alkanes contain only single carbon-carbon bonds. Alkenes contain at least one carbon-carbon double bond. Alkynes contain at least one carbon-carbon triple bond. The parent name is based on the longest carbon chain.

Functional Groups

Alcohol: hydroxyl (-OH). **Aldehyde:** terminal carbonyl (-CHO). **Ketone:** internal carbonyl (C=O). **Carboxylic acid:** -COOH. **Ester:** -COO- between carbon groups.

Naming Example

CH₃-CH(OH)-CH₃ has a three-carbon chain and an -OH group on carbon 2, so its name is propan-2-ol (2-propanol).

Structure and Properties

Functional groups affect intermolecular forces. Molecules capable of hydrogen bonding generally have higher boiling points than similar-sized non-polar hydrocarbons.

Practice

- Name CH₃CH₂CH₂CH₃.
- Identify the functional group in CH₃COOH.
- Explain why ethanol has a higher boiling point than ethane.
- Draw a structural formula for butan-2-ol.